

AI in Brain Tumor Diagnosis

Classification and care

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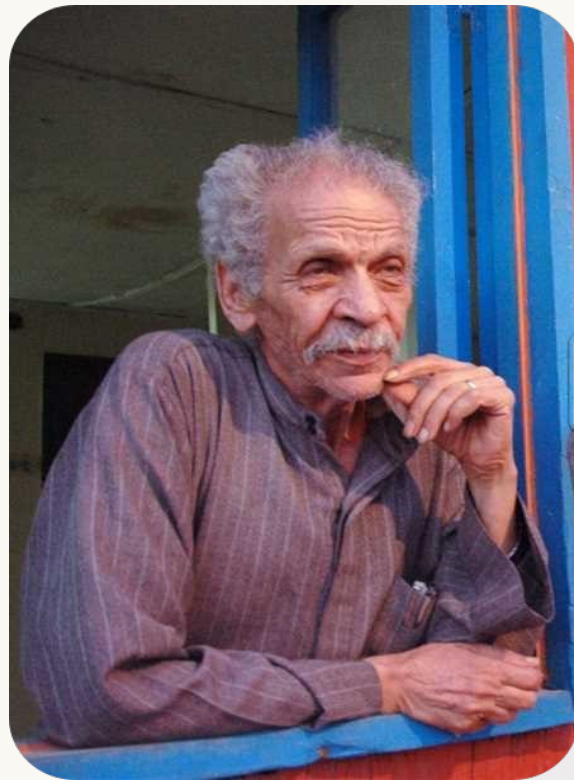
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PROBLEM

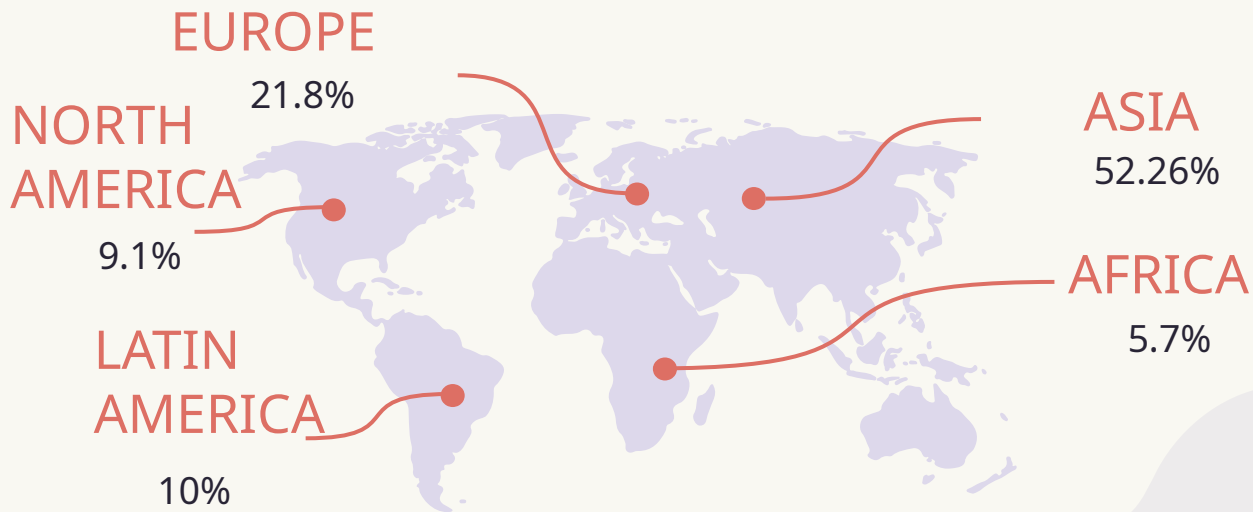


1M Affected

90,000 New/Y

19,000
Death/Y

8740
Egy-2025

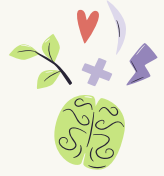




Early Detected
Can survive up to
20 years



Late Detected
Can survive up to
5 years





SOLUTION



OUR WORK



FOR EXPERTS:

DETECTION
AND
CLASSIFICATION

FOR PATIENTS:

MEDICAL
CHATBOT

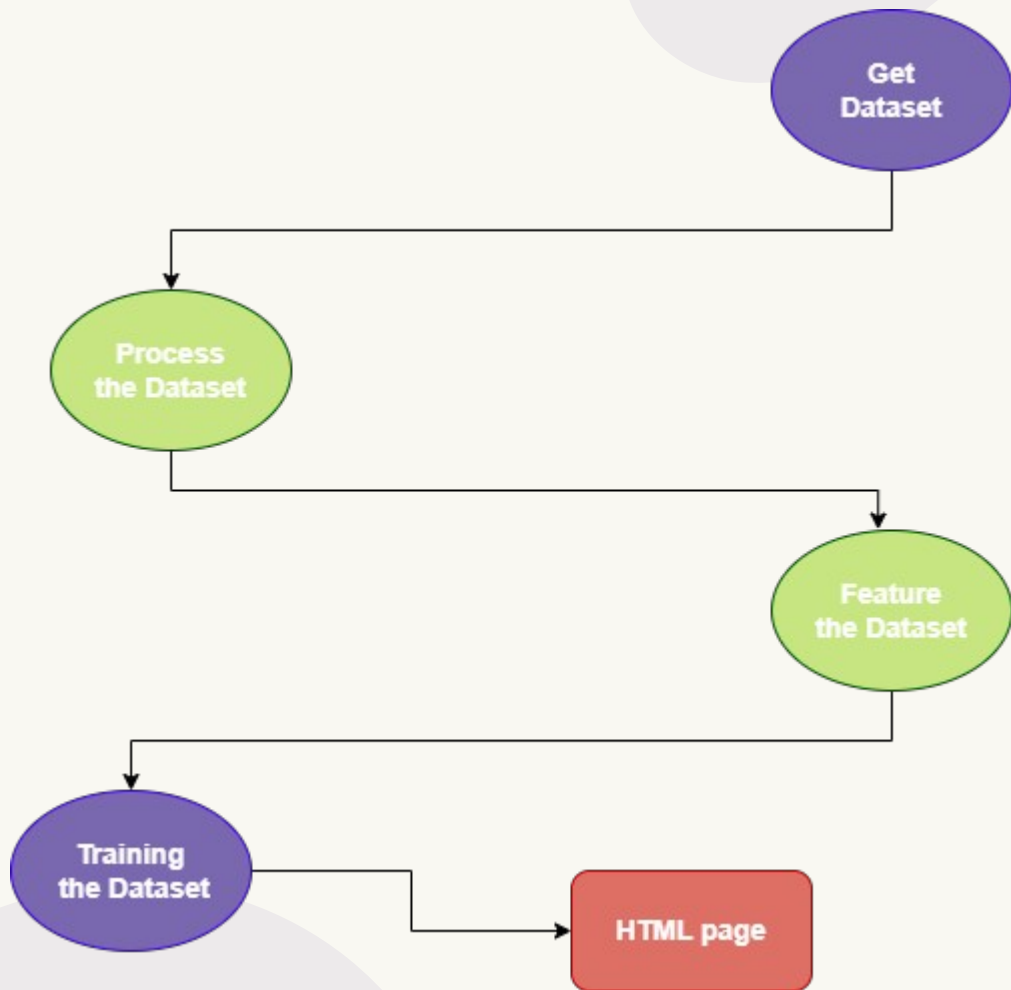
01

DETECTION AND CLASSIFICATION



Training:

- Data Collection
- Data processing
- Feature Extraction
- Model Selection
- Model Training
- Model Evaluation
- Model Testing
- Fine-Tuning and Optimization



WE USED:

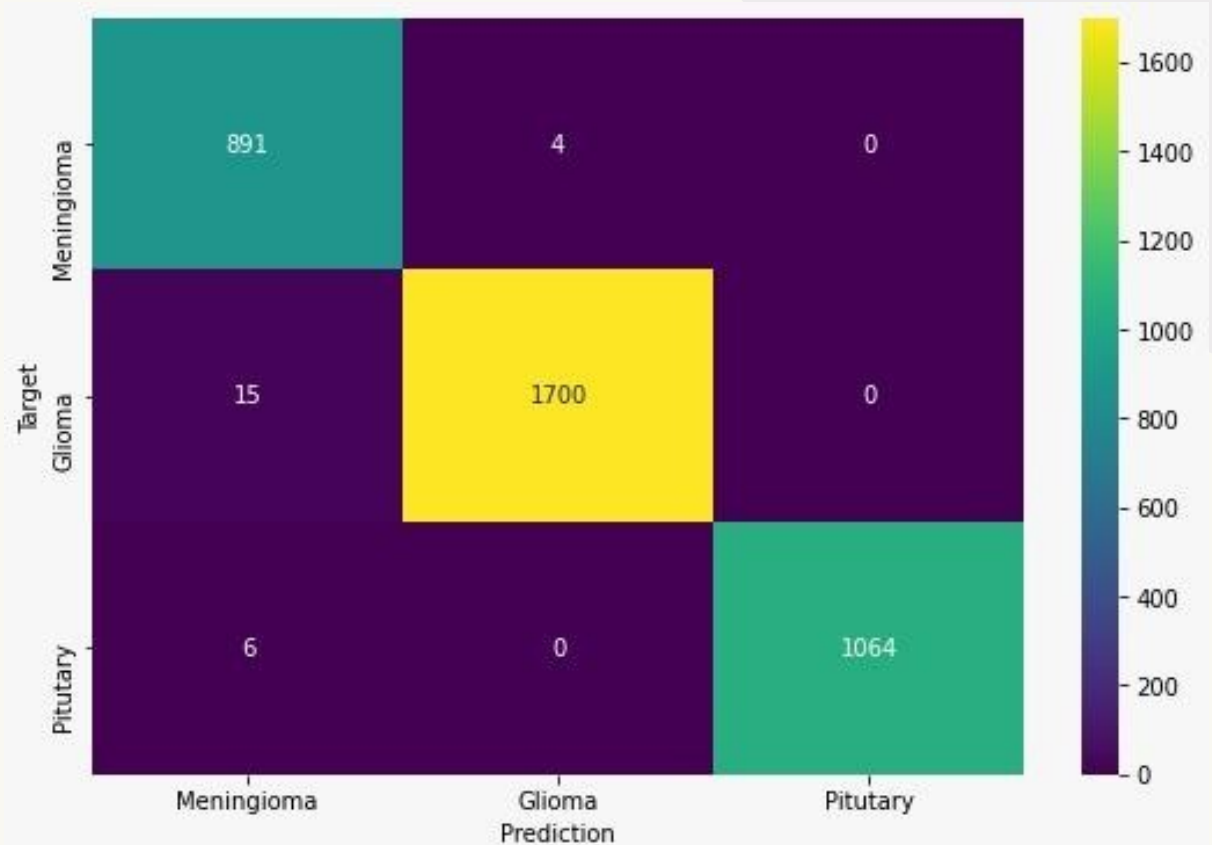
Some of the most important python libraries were used, such as:

- Os
- Requests
- Flask
- PIL
- TF
- Pytorch
- Pandas
- Scipy
- Matplotlib
- Numpy
- Werkzeug libraries



RESNET-50 MODEL

Heatmap of Confusion Matrix:



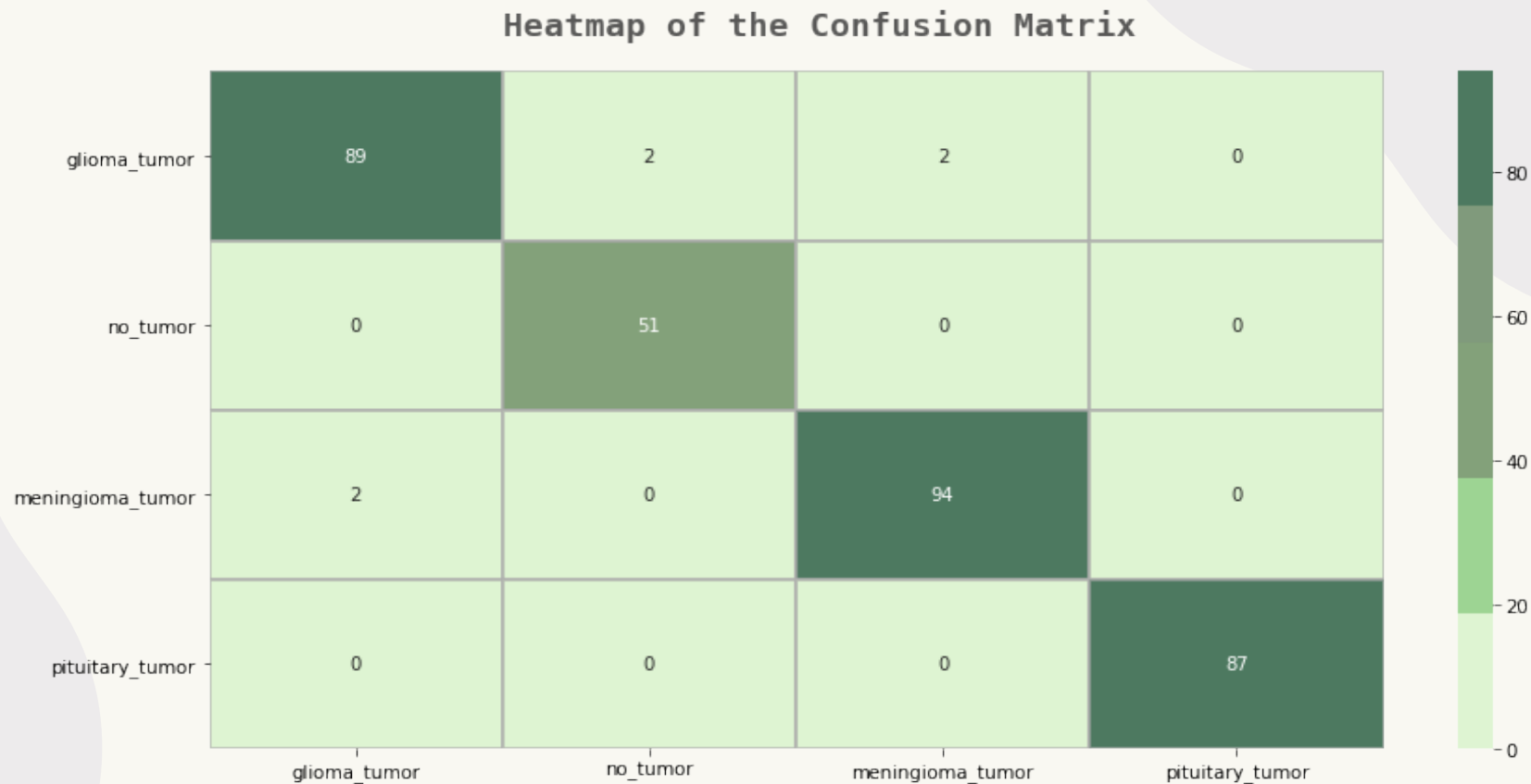


TENSORFLOW MODEL

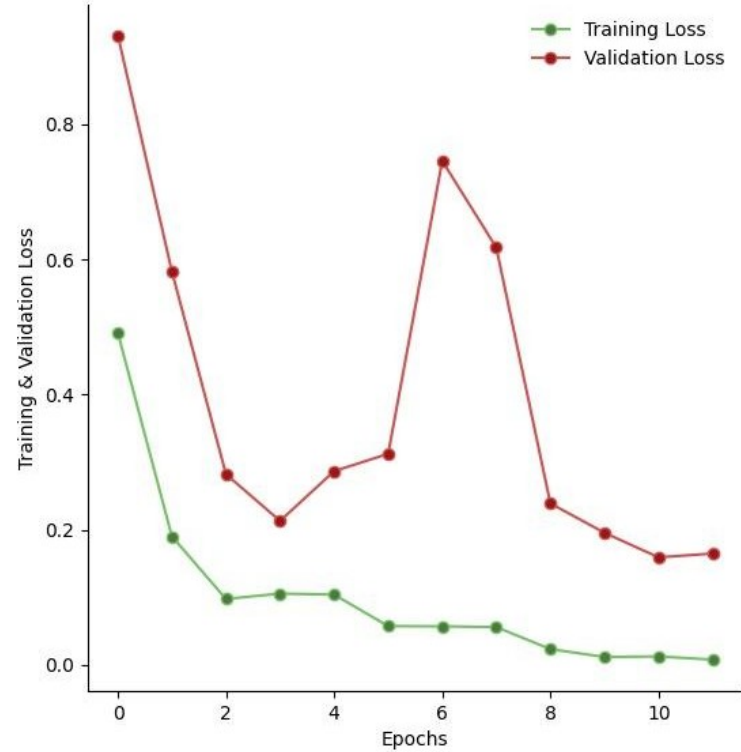
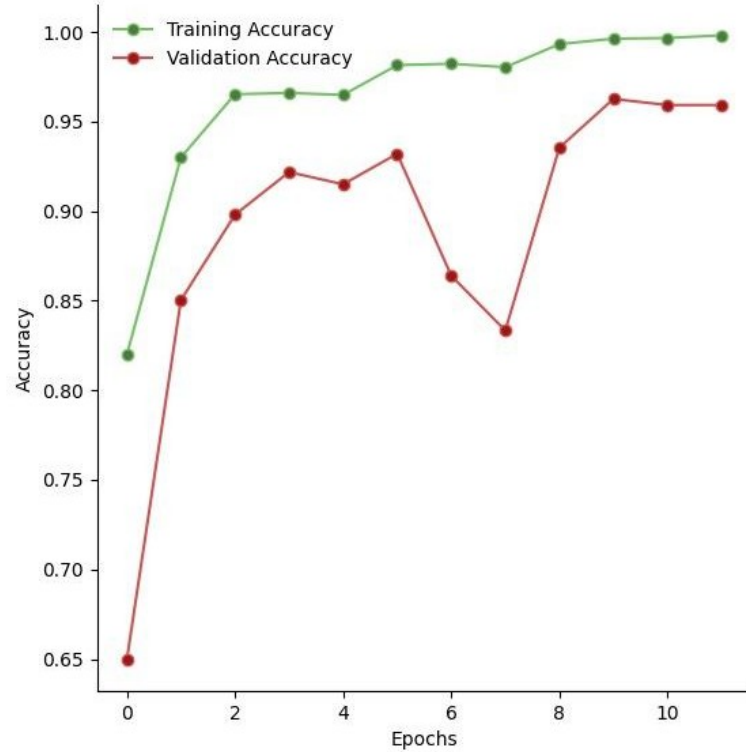
Classification Report

	precision	recall	f1-score	support
meningioma_tumor	0.98	0.96	0.97	93
no_tumor	0.96	1.00	0.98	51
pituitary_tumor	0.98	0.98	0.98	96
glioma	1.00	1.00	1.00	87
accuracy	-	-	0.98	327
Macro avg	0.98	0.98	0.98	327
Weighted avg	0.98	0.98	0.98	327

Heatmap of Confusion Matrix



Training VS Validation Performance



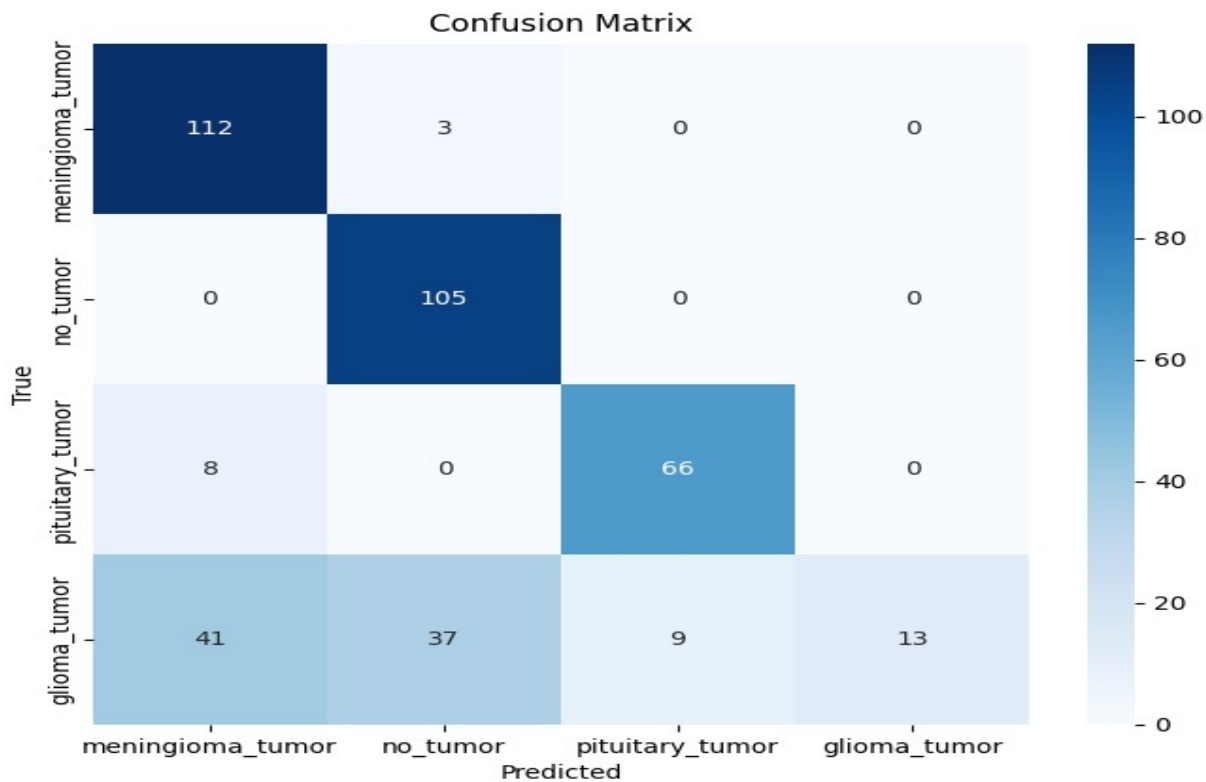


KAN MODEL

Classification Report

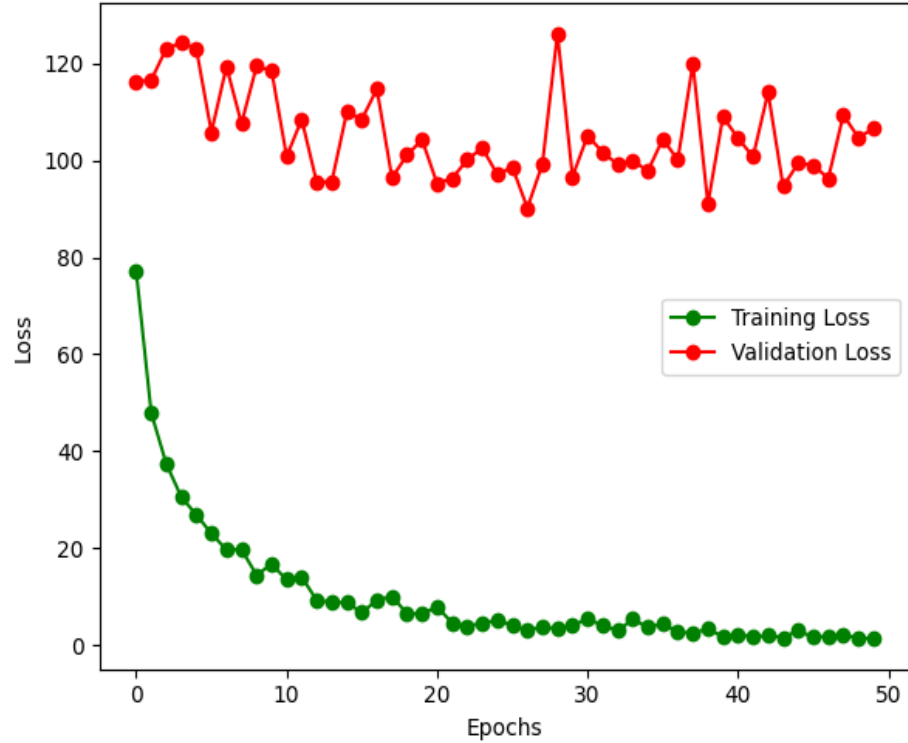
	precision	recall	f1-score	support
meningioma_tumor	0.70	0.97	0.81	115
no_tumor	0.72	1.00	0.84	105
pituitary_tumor	0.88	0.89	0.89	74
glioma	1.00	0.13	0.23	100
accuracy	-	-	0.75	394
Macro avg	0.82	0.75	0.69	394
Weighted avg	0.82	0.75	0.69	394

Heatmap of Confusion Matrix

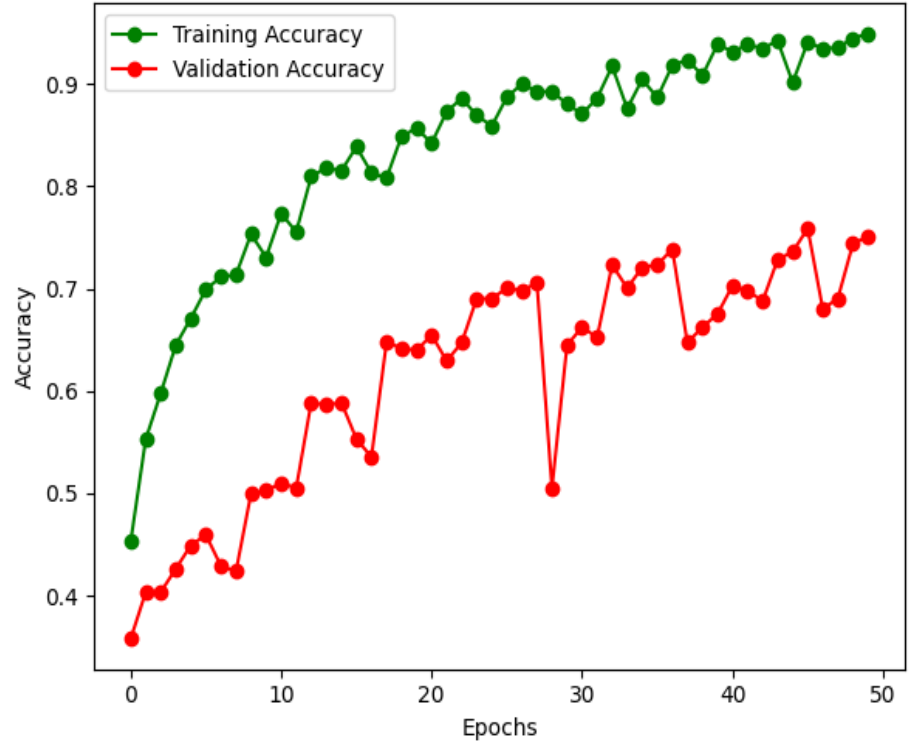


Training VS Validation Performance

Training & Validation Loss



Training & Validation Accuracy



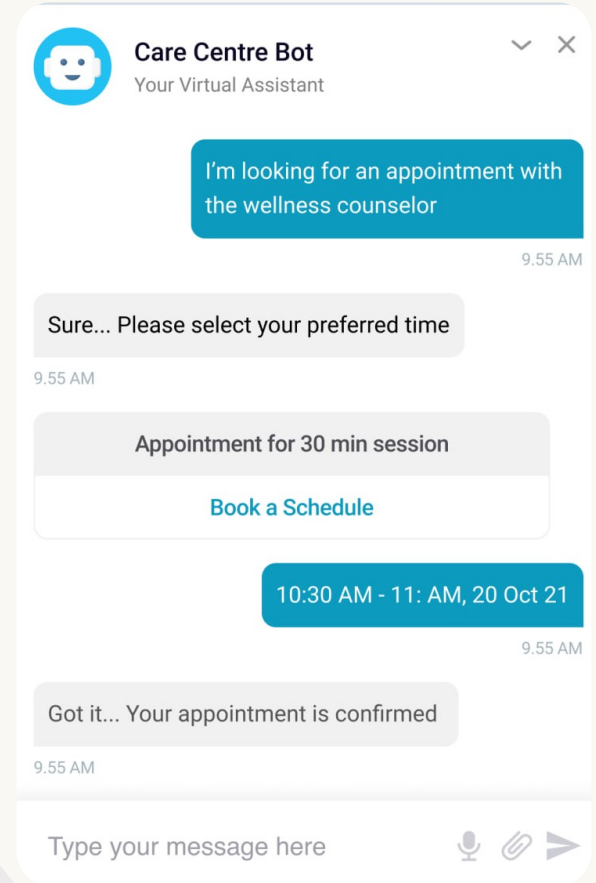
02

MEDICAL CHATBOT

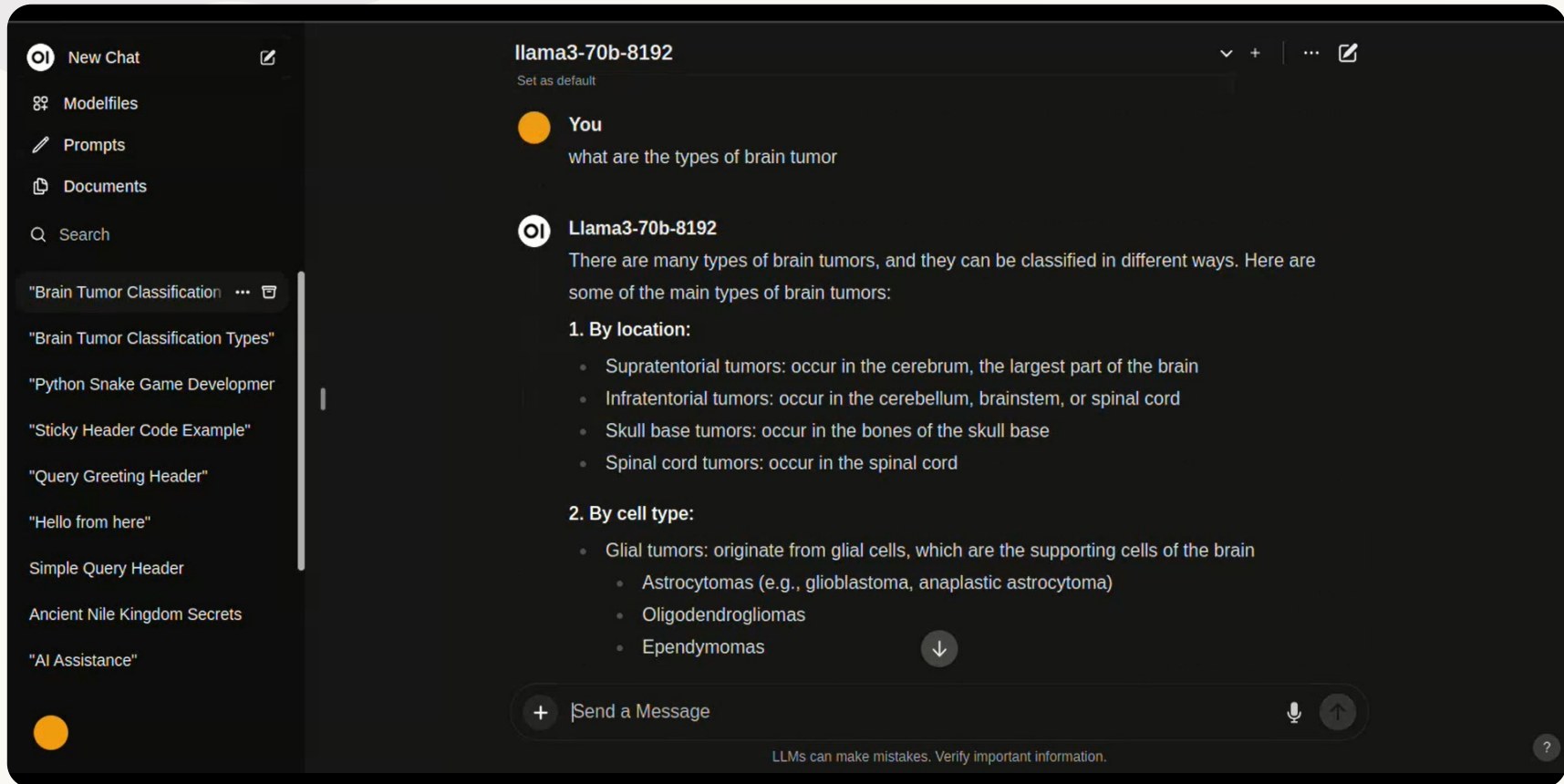


What is a Medical chatbot?

- Is a conversational AI-powered solution
- Designed to make healthcare more interactive and proactive.
- Chatbots were used also for firsthand symptom assessment.
- Healthcare chatbots act as the mediator between patients and their healthcare providers.

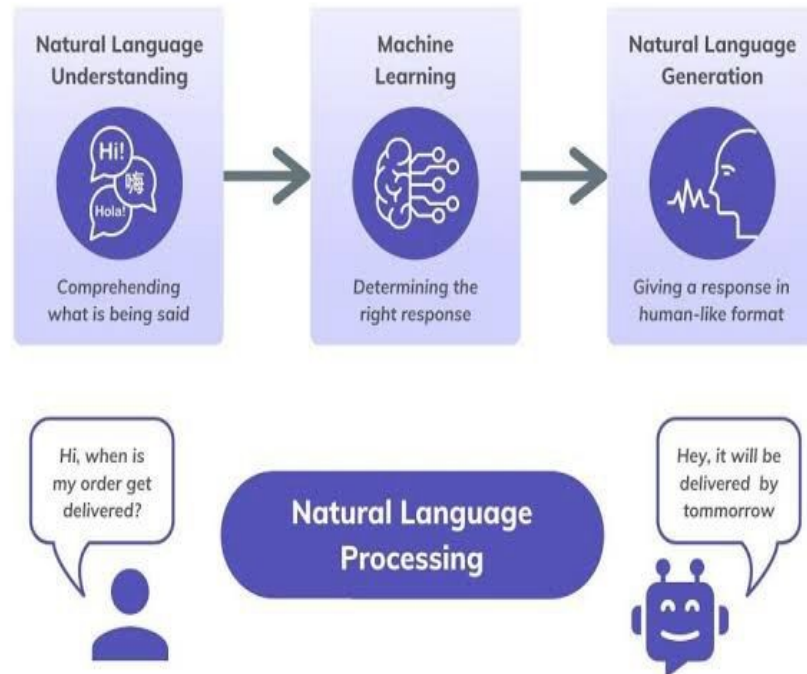


Our medical chatbot:



The relationship between NLP and CHATBOT:

- Understanding User Input
- Sentiment Analysis
- Language Translation
- Context Management
- Information Retrieval
- Named Entity Recognition



WE USED:

Some of the most important models and apps were used, such as:

- Ollama

- Docker

- Open WebUI

- Meta



Results





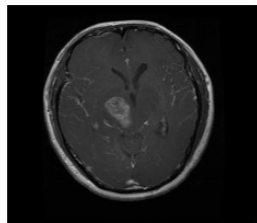
Classification Output

NeuroAI Detectives

Upload an Image

Classify

where the test image can be provided



	
Glioma	Glioma

[Go home](#)

This is how the pred page is displayed
(in this case it's Glioma, there are more types to classify)





Conclusion



How exactly is this project helpful?



1) Assisting healthcare professionals ,by:

- Enhancing their decision
- Speeding the classification process
- Decreasing human-error occurrence





2) Improved accessibility

3) Cost-Efficiency

4) Early Detection

5) Patients after-care using Medical Chatbot

THANKS!

